This week's exercises aim to prepare us for our midterm exam tomorrow.

0. Source Populate_Database_mynewpaltz.sql

I've sourced the file, let's examine the tables we've got:

CourseTable:

```
MariaDB [mynewpaltz]> select * from CourseTable;
+----------+-----------+--------------+-----------------------------+--------------+
| courseId | subjectId | courseNumber | title                       | numOfCredits |
+----------+-----------+--------------+-----------------------------+--------------+
|        1 | MATH      |          101 | Calculus I                  |            4 |
|        2 | MATH      |          102 | Calculus II                 |            4 |
|        3 | MATH      |          201 | Linear Algebra              |            3 |
|        4 | MATH      |          300 | Probability and Statistics  |            3 |
|        5 | MATH      |          405 | Abstract Algebra            |            3 |
|        6 | CS        |          101 | Intro to Computer Science   |            4 |
|        7 | CS        |          210 | Data Structures             |            3 |
|        8 | CS        |          315 | Operating Systems           |            4 |
|        9 | CS        |          402 | Database Management Systems |            3 |
|       10 | CS        |          450 | Artificial Intelligence     |            3 |
|       11 | ENGL      |          101 | English Composition I       |            3 |
|       12 | ENGL      |          102 | English Composition II      |            3 |
|       13 | ENGL      |          205 | World Literature            |            3 |
|       14 | ENGL      |          310 | Creative Writing            |            3 |
|       15 | ENGL      |          415 | Shakespeare                 |            3 |
|       16 | ENGR      |          100 | Intro to Engineering        |            2 |
|       17 | ENGR      |          201 | Statics                     |            3 |
|       18 | ENGR      |          202 | Dynamics                    |            3 |
|       19 | ENGR      |          305 | Thermodynamics              |            4 |
|       20 | ENGR      |          410 | Circuit Analysis            |            4 |
+----------+-----------+--------------+-----------------------------+--------------+
20 rows in set (0.000 sec)
```

StudentTable:

```
MariaDB [mynewpaltz]> select * from StudentTable;
+-----------+------------+------+-----------+---------+-----------+-----------------+---------+--------+
| ssn       | firstName  | mi   | lastName  | phone   | birthDate | street          | zipCode | deptId |
+-----------+------------+------+-----------+---------+-----------+-----------------+---------+--------+
| 100000001 | James      | A    | Smith     | 5550101 | 2002-05-15 | 101 Maple St    |   10001 | CS     |
| 100000002 | Mary       | B    | Johnson   | 5550102 | 2001-08-22 | 102 Oak Ave     |   10002 | MATH   |
| 100000003 | Robert     | C    | Williams  | 5550103 | 2003-01-10 | 103 Pine Rd     |   10003 | ENGL   |
| 100000004 | Patricia   | D    | Brown     | 5550104 | 2002-11-30 | 104 Elm St      |   10004 | ENGR   |
| 100000005 | John       | E    | Jones     | 5550105 | 2001-03-14 | 105 Cedar Ln    |   10005 | CS     |
| 100000006 | Jennifer   | F    | Garcia    | 5550106 | 2002-07-19 | 106 Birch Blvd  |   10006 | MATH   |
| 100000007 | Michael    | G    | Miller    | 5550107 | 2003-02-25 | 107 Walnut Way  |   10007 | ENGL   |
```

EnrollmentTable:

```
MariaDB [mynewpaltz]> select * from EnrollmentTable;
+-----------+----------+----------------+-------+
| ssn       | courseId | dateRegistered | grade |
+-----------+----------+----------------+-------+
| 100000001 |        1 | 2024-01-10     | A     |
| 100000001 |        6 | 2024-01-10     | B     |
| 100000001 |       11 | 2024-01-10     | A     |
| 100000002 |        2 | 2024-01-10     | B     |
| 100000002 |        7 | 2024-01-10     | C     |
| 100000002 |       12 | 2024-01-10     | B     |
```

1. List students by name and ssn who are taking 5 courses

To do this, I'll create a view `fiveOrMore` representing all of the ssns from EnrollmentTable that are taking 5 or more classes.

```
CREATE OR REPLACE VIEW fiveOrMore AS SELECT COUNT(*) AS numCourses, ssn FROM
EnrollmentTable GROUP BY ssn HAVING numCourses > 4;
```

fiveOrMore:

```
SELECT * FROM fiveOrMore;
+------------+-----------+
| numCourses | ssn       |
+------------+-----------+
|          5 | 100000041 |
|          5 | 100000042 |
|          5 | 100000043 |
|          5 | 100000044 |
|          5 | 100000045 |
|          5 | 100000047 |
|          5 | 100000048 |
|          5 | 100000049 |
|          5 | 100000050 |
|          5 | 100000052 |
|          5 | 100000053 |
|          5 | 100000054 |
|          5 | 100000055 |
|          5 | 100000056 |
|          5 | 100000058 |
|          5 | 100000059 |
|          5 | 100000060 |
+------------+-----------+
```

Now, I need to join this with the StudentTable to match ssns to names.

```
SELECT f.numCourses, f.ssn, s.firstName FROM fiveOrMore f JOIN StudentTable
s on f.ssn = s.ssn;
```

Output:

```
+------------+-----------+-----------+
| numCourses | ssn       | firstName |
+------------+-----------+-----------+
|          5 | 100000041 | Kevin     |
|          5 | 100000042 | Michelle  |
|          5 | 100000043 | Brian     |
|          5 | 100000044 | Carol     |
|          5 | 100000045 | Edward    |
|          5 | 100000047 | Ronald    |
|          5 | 100000048 | Melissa   |
|          5 | 100000049 | Timothy   |
|          5 | 100000050 | Deborah   |
|          5 | 100000052 | Stephanie |
|          5 | 100000053 | Jeffrey   |
|          5 | 100000054 | Rebecca   |
|          5 | 100000055 | Ryan      |
|          5 | 100000056 | Laura     |
|          5 | 100000058 | Sharon    |
|          5 | 100000059 | Gary      |
|          5 | 100000060 | Cynthia   |
+------------+-----------+-----------+
```

2. List students by name and ssn who are taking 4 courses

Just alter the view:

```
CREATE OR REPLACE VIEW four AS SELECT COUNT(*) AS numCourses, ssn FROM
EnrollmentTable GROUP BY ssn HAVI
NG numCourses = 4;
```

And the SELECT:

```
SELECT f.numCourses, f.ssn, s.name FROM four f JOIN StudentTable s on f.ssn
= s.ssn;
```

```
+------------+-----------+-----------+
| numCourses | ssn       | firstName |
```

```
+------------+-----------+-----------+
|          4 | 100000046 | Amanda    |
|          4 | 100000057 | Jacob     |
+------------+-----------+-----------+
```

3. List students by name and ssn who are taking 0 courses

```
SELECT * FROM StudentTable s WHERE s.ssn NOT IN (SELECT ssn FROM
EnrollmentTable);
```

Output:

```
+-----------+-----------+------+----------+---------+------------+---------------+---------+--------+
| ssn       | firstName | mi   | lastName | phone   | birthDate  | street        | zipCode | deptId |
+-----------+-----------+------+----------+---------+------------+---------------+---------+--------+
| 100000051 | Jason     | Y    | Gomez    | 5550151 | 2001-09-29 | 151 Lily Dr   |   10051 | ENGL   |
| 100000091 | Jack      | M    | Myers    | 5550191 | 2002-04-03 | 191 Millet Ln |   10091 | ENGL   |
+-----------+-----------+------+----------+---------+------------+---------------+---------+--------+
```

4. Give a count of 4 credit courses by subjectID (# of 4-credit courses in each subject).

```
SELECT COUNT(*), subjectId FROM CourseTable WHERE numOfCredits = 4 GROUP BY
subjectId;
```

```
+----------+-----------+
| COUNT(*) | subjectId |
+----------+-----------+
|        2 | CS        |
|        2 | ENGR      |
|        2 | MATH      |
+----------+-----------+
```

5. Give a count of 3 credit courses by subjectID

```
SELECT COUNT(*), subjectId FROM CourseTable WHERE numOfCredits = 3 GROUP BY
subjectId;
```

```
+----------+-----------+
| COUNT(*) | subjectId |
+----------+-----------+
|        3 | CS        |
|        5 | ENGL      |
|        2 | ENGR      |
```

```
|         3 | MATH      |
+----------+-----------+
```

Small change.

6. Are there any 5 credit courses? Are there any 1 credit courses?

```
MariaDB [mynewpaltz]> SELECT COUNT(*), subjectId FROM CourseTable WHERE
numOfCredits = 5 GROUP BY subjectId;
Empty set (0.001 sec)

MariaDB [mynewpaltz]> SELECT COUNT(*), subjectId FROM CourseTable WHERE
numOfCredits = 1 GROUP BY subjectId;
Empty set (0.001 sec)
```

No.

7. List all enrolled students by name and their grades in each enrolled course

Looks like a triple join here.

First, let's join EnrollmentTable to StudentTable in a view.

```
CREATE OR REPLACE VIEW es AS SELECT e.ssn,e.courseId, e.grade, s.firstName
FROM EnrollmentTable e JOIN StudentTable s ON e.ssn = s.ssn GROUP BY e.ssn;
```

```
SELECT * FROM es;
+-----------+----------+-------+-------------+
| ssn       | courseId | grade | firstName   |
+-----------+----------+-------+-------------+
| 100000001 |        1 | A     | James       |
| 100000002 |        2 | B     | Mary        |
| 100000003 |        3 | A     | Robert      |
| 100000004 |        4 | B     | Patricia    |

etc
```

Now, let's join our view to CourseTable.

```
SELECT c.title, es.firstName, es.grade FROM CourseTable c JOIN es ON
```

```
c.courseId = es.courseId;
```

Output:

```
+-----------------------------+-------------+-------+
| title                       | firstName   | grade |
+-----------------------------+-------------+-------+
| Calculus I                  | Tyler       | A     |
| Calculus I                  | Janet       | A     |
| Calculus I                  | Frank       | C     |
| Calculus I                  | Anna        | B     |
| Calculus I                  | Cynthia     | A     |
| Calculus I                  | Timothy     | A     |
| Calculus I                  | Edward      | B     |
| Calculus I                  | Kevin       | A     |
| Calculus I                  | Charles     | A     |
```

8. For all courses identified by courseId of 10, list # of students enrolled, course number, & subjectId

```
SELECT * FROM CourseTable where CourseID = 10;
```

```
+----------+-----------+--------------+-------------------------+-----------
---+
| courseId | subjectId | courseNumber | title                   |
numOfCredits |
+----------+-----------+--------------+-------------------------+-----------
---+
|       10 | CS        |          450 | Artificial Intelligence |
3 |
+----------+-----------+--------------+-------------------------+-----------
---+
```

Now that we see course with ID 10, let's get # of students enrolled. I'll use a nested SELECT:

```
SELECT COUNT(*) AS enrolledStu, e.courseID, a.subjectId, a.title FROM
EnrollmentTable e JOIN (SELECT * FROM CourseTable WHERE CourseID = 10) AS a
ON e.courseId = a.courseId;
```

output:

```
+-------------+----------+-----------+-------------------------+
| enrolledStu | courseID | subjectId | title                   |
+-------------+----------+-----------+-------------------------+
|          13 |       10 | CS        | Artificial Intelligence |
+-------------+----------+-----------+-------------------------+
```

9. How many students are in each course (identified uniquely by courseID)?

Simple nested SELECT:

```
SELECT q.enrolledStu, c.title, q.courseID FROM CourseTable c JOIN (SELECT
COUNT(*) AS enrolledStu, courseID
FROM EnrollmentTable GROUP BY courseID) AS q ON q.courseID = c.courseID;
```

```
+-------------+----------------------------+----------+
| enrolledStu | title                      | courseID |
+-------------+----------------------------+----------+
|          13 | Calculus I                 |        1 |
|          13 | Calculus II                |        2 |
|          12 | Linear Algebra             |        3 |
|          14 | Probability and Statistics |        4 |
|          13 | Abstract Algebra           |        5 |
|          11 | Intro to Computer Science  |        6 |
|          13 | Data Structures            |        7 |
|          13 | Operating Systems          |        8 |
|          12 | Database Management Systems |        9 |
etc
```

10. For each student enrolled, list the first name, last name, and number of credits that student was taking.

```
SELECT s.firstName, s.lastName, sum(c.numOfCredits) AS TotalCreditsTaken
FROM StudentTable s JOIN EnrollmentTable e ON s.ssn = e.ssn JOIN CourseTable
c ON e.CourseId=c.CourseId group by s.ssn ORDER BY TotalCreditsTaken desc;
```

```
+-------------+------------+-------------------+
| firstName   | lastName   | TotalCreditsTaken |
+-------------+------------+-------------------+
| Michelle    | Adams      |                17 |
| Kevin       | Green      |                17 |
| Deborah     | Roberts    |                17 |
```

```
| Laura       | Parker     |                 17 |
| Carol       | Baker      |                 16 |
| Jeffrey     | Evans      |                 16 |
| Rebecca     | Turner     |                 16 |
etc
```